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Multiprogram Control — Current Thinking of W. Heising, J. Batchelder and E. F. Codd

A. Some Basic Objectives

1. Use of the CPU and input-output units is to be shared by two or more programs in such a way that the total execution time for these programs taken collectively has a high degree of probability of being less than the total execution time if the programs were executed one after another.

2. When any problem program is run by itself its input-output operations will be overlapped with CPU operations to the degree permitted by that program and by the machine.

3. Inter-problem priorities will be controlled exclusively by the supervisory program and cannot be affected by problem programs. Intra-problem priorities on the other hand may be specified and changed whenever necessary by the problem program.

4. Maximum utilization of the CPU and input-output units will be attempted within the framework of existing program priorities.

5. When programming a problem which may be run under multiprogram control, the programmer need have no knowledge of the nature of any other problems with which it may be concurrently executed.

6. The amount of additional information required within or along with a problem program if it is to be run under multiprogram control should be kept to a minimum.

7. No problem program should be allowed to introduce error into any other problem program or into the supervisory program.

B. Some Principles of Implementation

1. In order to comply with the "non-interference" objective (#7) problem programs will issue I/O instructions to the supervisory program instead of to the machine.

2. All interrupts will be handled by the supervisory program.

3. Any changes required by the problem program in the interrupt table, mask, indicators, interrupt address, and the enabled-disabled status of the interrupt mechanism must be requested of the supervisory program. The supervisory program will refuse a request which could interfere with any other program or with the supervisory program itself, and will then notify the problem program of this refusal.

4. The normal mode in which the supervisory program will handle input-output operations involves masking all I/O interrupts ON.

5. Whenever control is passed to, or taken by, the supervisory program, existing priorities will be examined to assist in determining what should be done next. A possible exception to this is the so-called single instruction fix-up.

C. Supervisory Tables for CPU Activity

1. Active Programs

An entry consists of:

Identification of problem program
Inter-problem priority
Stream bit

The stream bit indicates whether or not an instruction stream belonging to this program is waiting to be executed. It is possible that all CPU activity for a particular program is suspended until one or more I/O operations are completed. When the stream bit is ON, it indicates that the next table should be searched. Its status will change from time to time.

2. Permitted Streams

An entry consists of:

Identification of problem program
Intra-problem priority
Resume bit
Resume address
Location and identification of dumped quantities.

The resume bit permits or inhibits resumption of execution of this stream. Its status and also the resume address will change from time to time.

D. Supervisory Tables for I/O Activity

1. Not Yet Issued

An entry consists of:

Identification of problem program
Location of instruction
Instruction
Location for recording final status
Issuing priority

Those I/O instructions which cannot be executed immediately are stacked up in this table. Every opportunity is seized to initiate the execution of such instructions. When it is known that a waiting I/O instruction can be executed and its priority is sufficient, then its table entry is moved down to the next table.

2. Occupying Channel

Entries in this table have the same form as entries in the Not Yet Issued table except that issuing priority is omitted. There is precisely one entry per exchange channel, and the entries are permanently kept in channel number order. When a channel is released the entry associated with that channel is replaced with an indication that the channel is free. The completion of certain I/O operations (for example, rewind tape) is not normally coincident with release of the channel. In such cases

the I/O unit carries on to the end of its operation independently and finally provides a Unit Signal. For I/O operations of this kind the following table is required and at the time the channel is released the corresponding entry is moved down to this table.

3. Channel Released, Execution Not Completed

Entries in this table have the same form as entries in the Not Yet Issued table except that issuing priority is omitted.

E. Remarks on I/O Supervision

1. Not Yet Issued Bit

This bit indicates whether or not there exist I/O instructions in the Not Yet Issued table. If this bit is on when supervisory program has obtained control from a problem program, then at the earliest possible moment it takes this opportunity to issue as many of the "not yet issued" instructions as possible.

2. Symbolic Treatment of Unit Addresses

Whatever form the programmer originally used to identify the tape units, card machines, etc, required in his problem, the supervisory program is the final arbiter as to which physical I/O units are actually called for as a program is executed. The translation of a unit address as specified by the programmer into the actual unit address to be used is implemented by means of an I/O assignment table. It should be observed that an I/O unit may be assigned to more than one program at a time. Precautions will be necessary in the supervisory program to ensure non-interference, particularly when the I/O unit shared by two or more programs operates with an erasable medium (tape, disk) and more than one program calls for writing on that medium.

3. Termination of an I/O Operation

No matter how an I/O operation is terminated – whether it be normal completion, an end-of-file condition or an error – the final status is recorded in the location specified by the problem program at the time the operation was requested.

In this way the problem program may decide what action is appropriate. This does not preclude certain standard error procedures being available as part of the supervisory program. The choice of using them, however, is left up to the problem program.

The table entry for a terminated I/O operation is struck from the record after the final status has been transmitted.

F. Signals from Problem Programs to the Supervisory Program

REQUEST I/O

Specified at earliest point in a problem program at which it is known that an I/O operation is required to be executed.

1. I/O operation
2. Unit Address
3. Location of first ECW
4. Location for recording final status

WAIT

Specified at a point in a problem program beyond which no progress can be made until a certain I/O operation is completed.

1. Address for resuming
2. Identification of I/O op for which completion is awaited

This signal should not be confused with the built-in instruction SET AND WAIT.

WAIT ANY

Specified at a point in a problem program beyond which no progress can be made until any I/O operation belonging to this program is completed.

1. Address for resuming

REQUEST STATUS

Specified whenever a problem program requires information concerning an I/O operation which it has requested. The supervisory program determines first of all in which of its tables the specified operation is recorded. Only if it is in the Occupying Channel table does it consult the Exchange memory with a COPY ECW instruction.

1. Identification of I/O operation
2. Location for recording status

CANCEL I/O

This signal causes the supervisory program to terminate the specified operation if it has been started or void its issuance if it has not yet been started. Status information is supplied to the problem program.

1. Identification of I/O operation to be cancelled
2. Location for recording status at time of cancellation

CANCEL ALL I/O

All I/O operations for the particular program which sends this signal are terminated or voided.

1. Location of bit to be set ON at completion of all cancellations

SET MASK WORD

This request is used when a problem program requires many positions of the interrupt mask to be set. The request may be rejected by the supervisory program.

1. Address of new mask word
2. Address of super mask word
3. Location for recording reject bit

SET MASK BIT

Same as SET MASK WORD except that only one bit in the mask may be changed.

1. Bit address of bit to be changed
2. Status (ON or OFF) for this bit
3. Location for recording reject bit

REQUEST STREAM

Specified in a problem program when a new stream of instructions is to be executed on a time-shared basis with existing streams belonging to this problem and to other problems.

1. Address of start of stream
2. Identification of stream
3. Intra-problem priority
4. Location for recording final status.

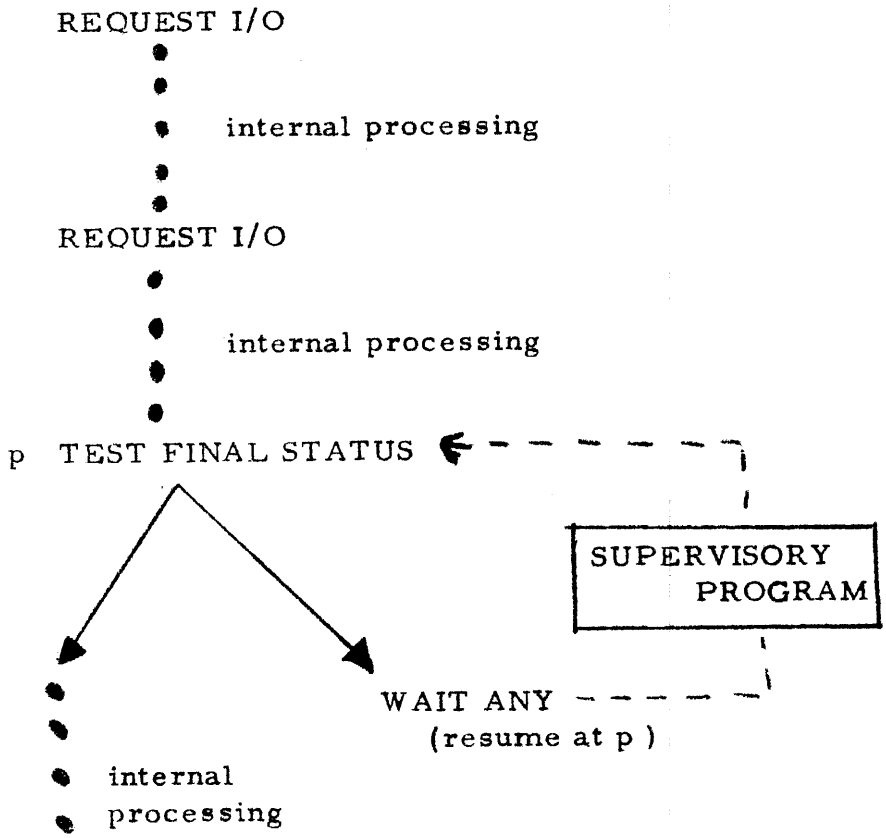
The supervisory program makes an entry in the Permitted Streams table and sets the Stream Bit on.

TERMINATE STREAM

Causes an entry in the Permitted Streams table to be erased. Prior to erasure the resume address is recorded in the location specified for the final status. A stream may terminate itself or any other stream belonging to the same problem program. The supervisory program will reject any attempt to terminate a stream that belongs to some other problem program or for which there is no entry in the Permitted Streams table.

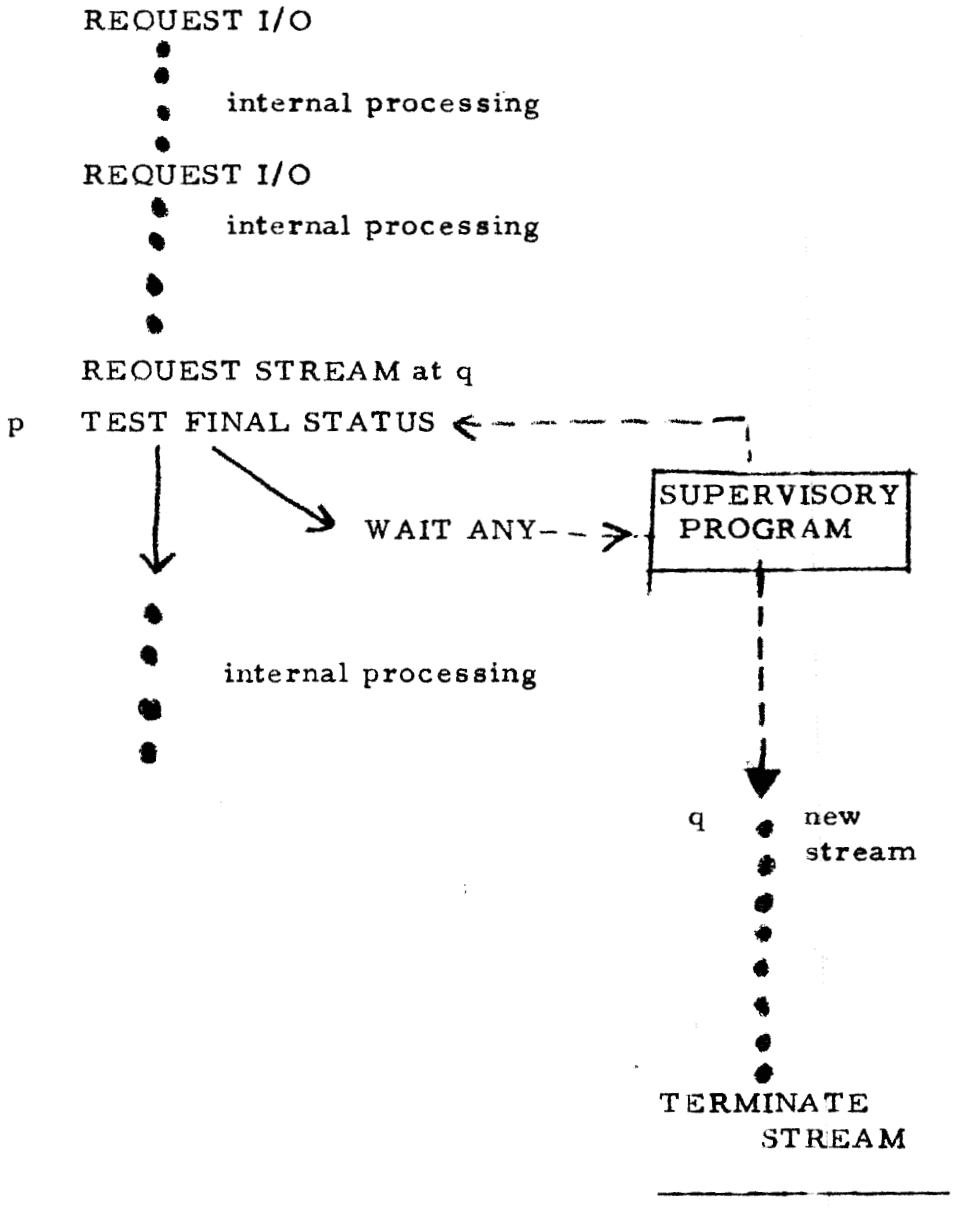
1. Identification of stream
2. Location for recording reject bit

G. Example of WAIT ANY in a Problem Program



The final status test may require that a particular combination of I/O operations be normally completed. In this way the necessary and sufficient conditions for proceeding are defined in the problem program.

H. Example of REQUEST STREAM and TERMINATE STREAM



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